BRUNVOLL

Technical Report

Propulsion Optimiser – Fuel Saving Potential

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1 Introduction

This report presents fuel saving results from Brunvoll's voyage fuel simulation tool, developed to quantify the potential of the **Propulsion Optimiser** – our smart propeller control system. The analysis also evaluates the combined benefit when the Propulsion Optimiser is paired with **Flettner rotor wind-assist**, as the two technologies are complementary: the Optimiser improves engine efficiency at every operating point, while the Flettner rotor reduces the thrust demand itself by harnessing wind energy. Together, they address different parts of the propulsion energy chain and their savings add up.

- **The Propulsion Optimiser** (referred to as "Optimiser" in the rest of this report) – finds the most fuel-efficient propeller pitch/RPM combination for any given condition, replacing the standard (factory-default) propeller control schedule.
- **Flettner rotor wind-assist** – a spinning cylinder mounted on deck that generates forward thrust via the Magnus effect, reducing the load on the main engine.

To evaluate these technologies under realistic conditions, we simulate a full year (2024) of repeated voyages on a representative North Sea route (Rotterdam–Gothenburg, 509 nm), using actual historical weather data. Each voyage experiences the real waves and winds that occurred on that date, giving a comprehensive picture of fuel savings across all seasons and weather conditions.

1.1 What We Compare

The simulation runs every voyage under four configurations:

Configuration	Description
Standard control, no rotor	Factory-default propeller schedule, no wind-assist – the baseline
Standard control + Flettner rotor	Factory schedule with wind-assist thrust from the rotor
Optimiser, no rotor	Fuel-optimised pitch/RPM selection, no wind-assist
Optimiser + Flettner rotor	Fuel-optimised pitch/RPM combined with wind-assist – the full-benefit case

Fuel savings are always measured relative to the baseline (standard control, no rotor), and are decomposed into:

- **Propeller optimisation saving** – the Optimiser's benefit from smarter pitch/RPM selection, independent of weather.
- **Wind-assist saving** – the additional benefit from the Flettner rotor, which grows with wind speed.

2 How the Simulation Works

2.1 Hull Resistance and Propulsion

The simulation models the vessel's propulsion chain from hull resistance through to engine fuel consumption:

- **Calm-water resistance** is based on the vessel's service prediction data (model test results), covering speeds from 8 to 15.5 knots.

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- **Wave added resistance** is computed using hydrodynamic transfer functions specific to the hull shape, integrated over the wave spectrum at each time step. In simple terms: the tool calculates how much extra drag the waves create based on their height, period, and direction relative to the vessel.
- **Wind resistance on the hull** follows the established Blendermann (1994) method, accounting for wind speed and direction relative to the vessel. Headwinds add resistance; following winds reduce it.
- **Hull and propeller fouling** can be included as time-dependent roughness growth. The baseline results shown here assume clean hull and propeller conditions.

2.2 Flettner Rotor Wind-Assist

The Flettner rotor is a spinning vertical cylinder ($28\,\mathrm{m}$ tall, $4\,\mathrm{m}$ diameter) that exploits the Magnus effect: when wind flows past the spinning rotor, it generates a sideways force – much like a spinning ball curves in flight. The forward-pointing component of this force acts as free thrust that supplements the propeller.

The key equation governing the rotor's thrust is:

$$F_{\text{rotor}} = C_L(\text{SR}) \cdot \tfrac{1}{2}\,\rho_{\text{air}}\,V_{\text{app}}^2 \cdot D \cdot H \tag{1}$$

where C_L is the lift coefficient (dependent on the spin ratio SR – how fast the rotor spins relative to the wind speed), V_{app} is the apparent wind speed, and $D \times H$ is the rotor's projected area. The rotor's own motor power consumption is subtracted to give the net fuel benefit.

The rotor produces the most benefit in beam winds (wind from the side) and provides diminishing returns in headwinds or following winds. Its effectiveness increases with wind speed.

2.3 Standard vs Optimiser Propeller Control

A controllable-pitch propeller (CPP) has two adjustment parameters: blade pitch and shaft RPM. The **standard control** uses a fixed schedule that links these two parameters together – a single lever controls both, following a pre-set curve.

The **Optimiser** breaks this rigid link. For any given thrust requirement, it searches the full range of pitch and RPM combinations to find the one that minimises fuel consumption, while respecting the engine's operating limits. The saving comes from running the engine at a more efficient load point – typically lower RPM and higher pitch than the standard schedule would dictate.

2.4 Weather Data

The simulation uses wave and wind data from NORA3, the third-generation Norwegian Reanalysis Archive produced by MET Norway. This is a high-quality historical weather reconstruction that provides complete, gap-free wave and wind fields across the North Sea at approximately $3\,\mathrm{km}$ spatial resolution and hourly time steps.

Unlike isolated buoy measurements, this dataset covers the entire route with consistent quality, ensuring that every simulated voyage encounters realistic weather conditions. The archive spans from 1993 to the present, enabling multi-year studies if needed.

3 Simulation Setup

The study simulates a full year of round-trip voyages on the route shown in Figure 3-1.

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Parameter	Value	Unit
Route	509	nm (one-way)
Transit speed	10	kn
Transit time (one-way)	50.9	hours
Mode	Round-trip	
Flettner rotor	28 m tall, 4 m dia	
Hull condition	Clean (no fouling)	
Port/idle time	15	%
Fuel price (MGO)	650	EUR/tonne
Propeller diameter	4.80	m
Main engine	MAN L27/38	2920 kW

Each voyage departs daily, sails the outbound leg, and returns immediately – experiencing different weather on each leg. Round-trip simulation eliminates any directional wind bias (e.g.\ prevailing westerlies favouring one direction). With 15% assumed port/idle time, the vessel completes approximately 73 round-trip voyages per year.

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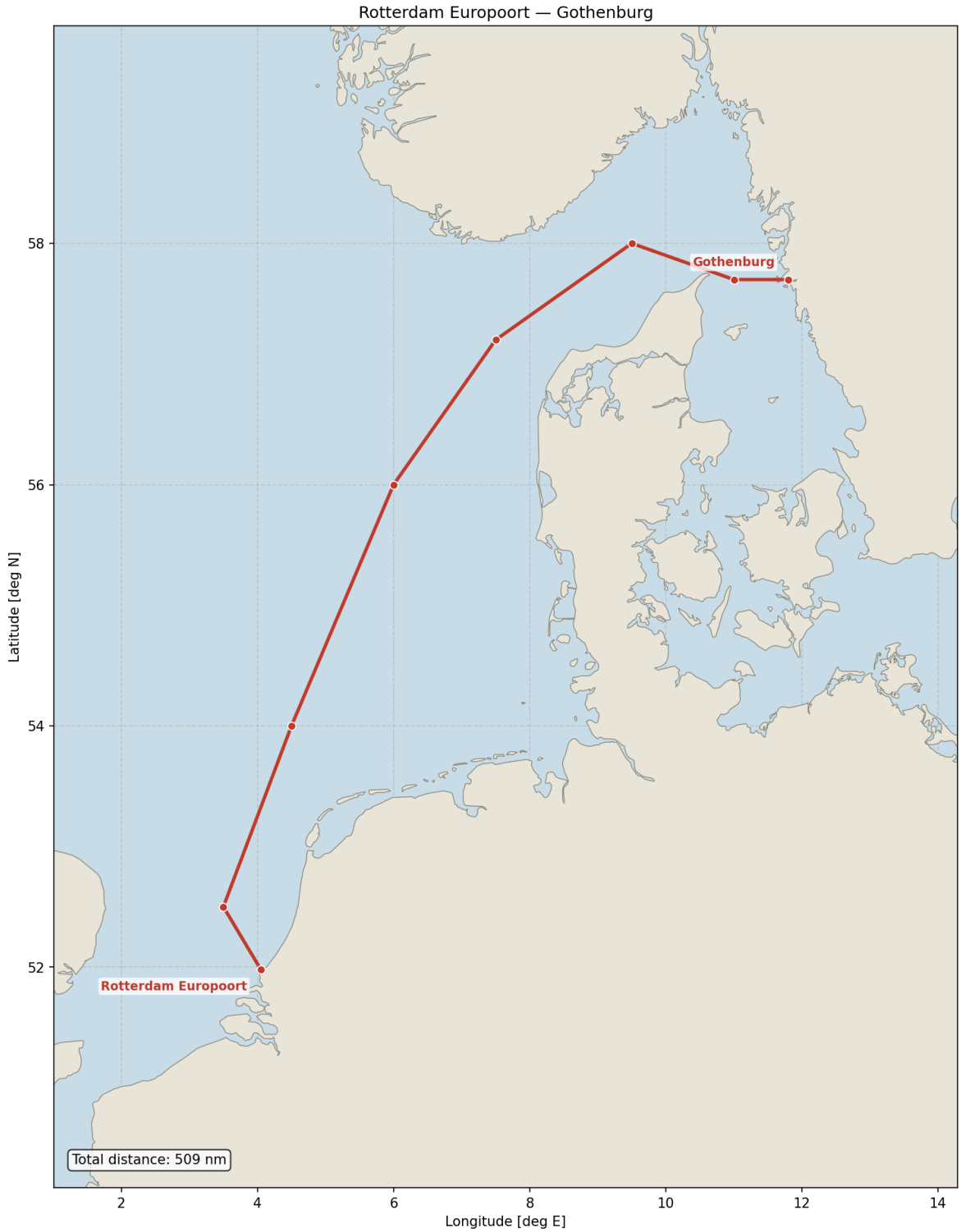

Figure 3-1: Rotterdam–Gothenburg route (509 nm one-way) with waypoints.

4 Results

362 round-trip voyages were successfully simulated across the full year 2024.

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4.1 Per-Voyage Fuel Consumption

Metric	Mean	Median	P10	P90	Min	Max
Standard, no rotor [kg/voyage]	14502	14392	13297	16200	10545	19300
Optimiser + rotor [kg/voyage]	12577	12620	10614	14417	8158	18030
Propeller optimisation saving [kg]	834	869	721	914	369	928
Wind-assist saving [kg]	1091	993	280	2091	-9	3527
Propeller optimisation saving	5.8%	6.0%	4.5%	6.6%	2.8%	7.0%
Wind-assist saving	7.7%	6.8%	1.9%	14.7%	-0.1%	25.0%

Key observations:

- The **propeller optimisation** delivers a consistent 3–7% saving on every voyage, regardless of weather.
- The **wind-assist saving** varies from near zero in calm conditions to over 25% on the windiest voyages, reflecting its weather dependence.
- On the single best voyage, the Flettner rotor alone saved 3.5 tonnes of fuel.

The savings are additive: the combined benefit equals the propeller optimisation saving plus the wind-assist saving.

Figure 4-1 shows how the savings evolve over the year, alongside the sea conditions that drive them.

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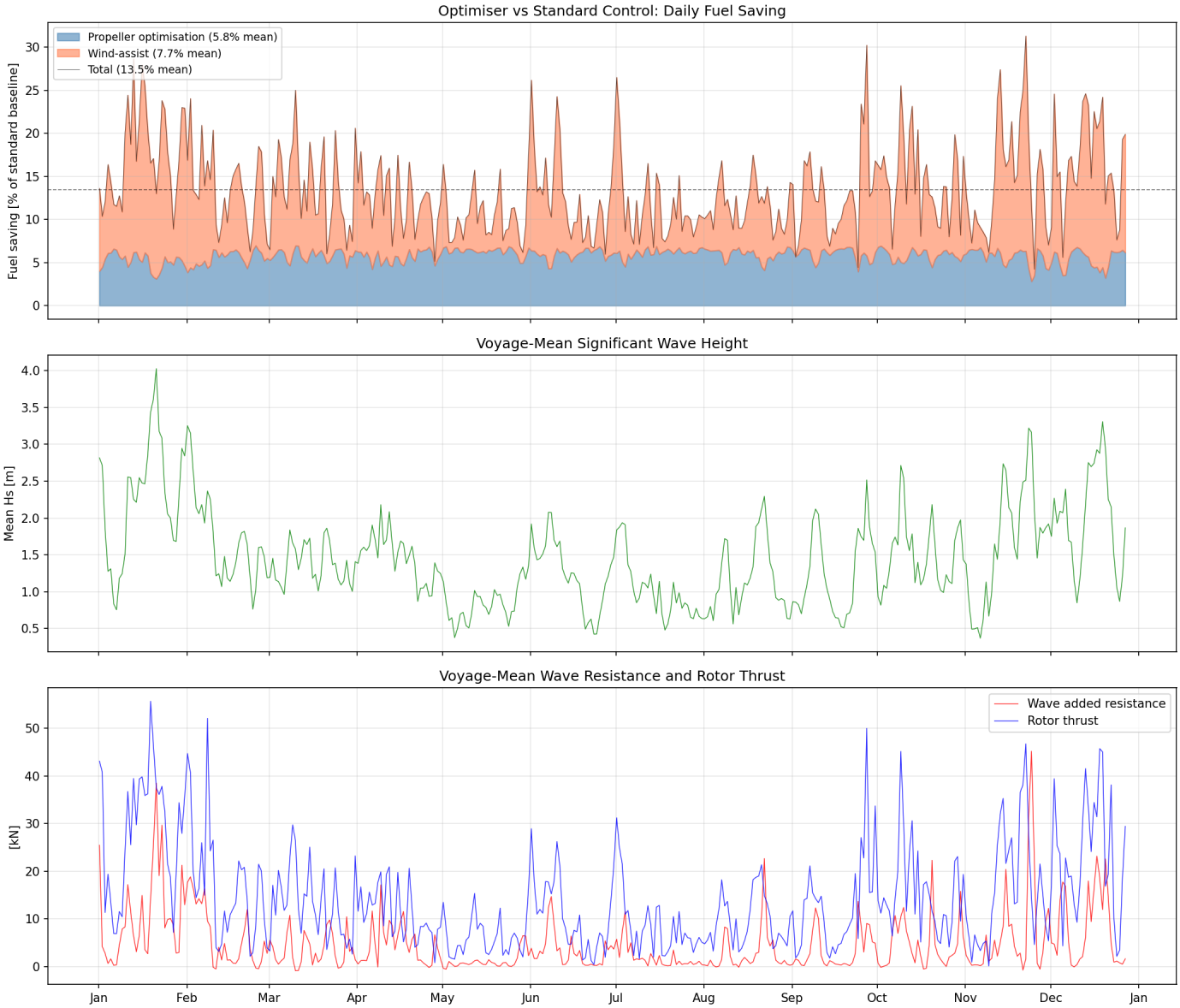

Figure 4-1: Annual results per voyage. Top: propeller optimisation and wind-assist fuel savings (stacked). Middle: significant wave height. Bottom: wave resistance, wind resistance, and Flettner rotor thrust.

4.2 Weather Correlation

Figure 4-2 shows how the fuel saving on each voyage correlates with wave height and wind speed. The trend is clear: windier conditions mean larger savings from the Flettner rotor.

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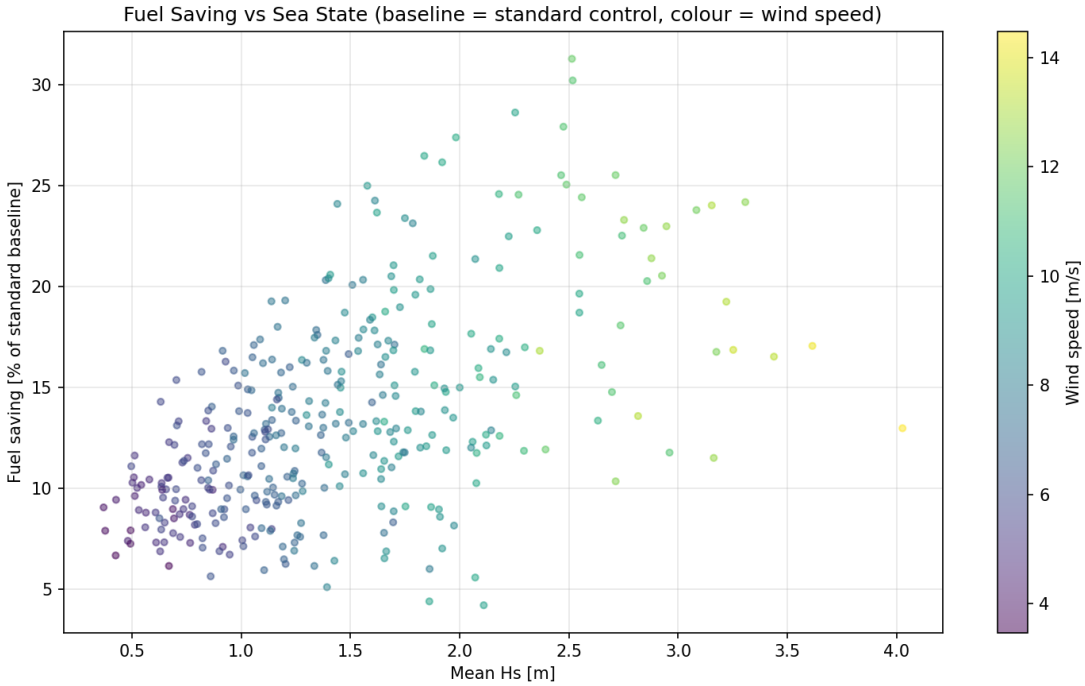

Figure 4-2: Per-voyage fuel saving vs mean wave height, coloured by mean wind speed. Windier conditions produce larger savings through the Flettner rotor.

4.3 Annualised Fuel Consumption

Configuration	Fuel [tonnes/year]
Standard control, no rotor	1060.5
Standard control + Flettner rotor	973.7
Optimiser, no rotor	999.5
Optimiser + Flettner rotor	919.7

Saving component	Tonnes/year	Percentage
Propeller optimisation	61.0	5.8%
Wind-assist (Flettner)	79.8	7.5%
Combined	**140.8**	**13.3%**

Figure 4-3 shows the distribution of per-voyage savings. The propeller optimisation is tightly clustered (consistent benefit), while the wind-assist saving shows a wider spread reflecting its dependence on weather.

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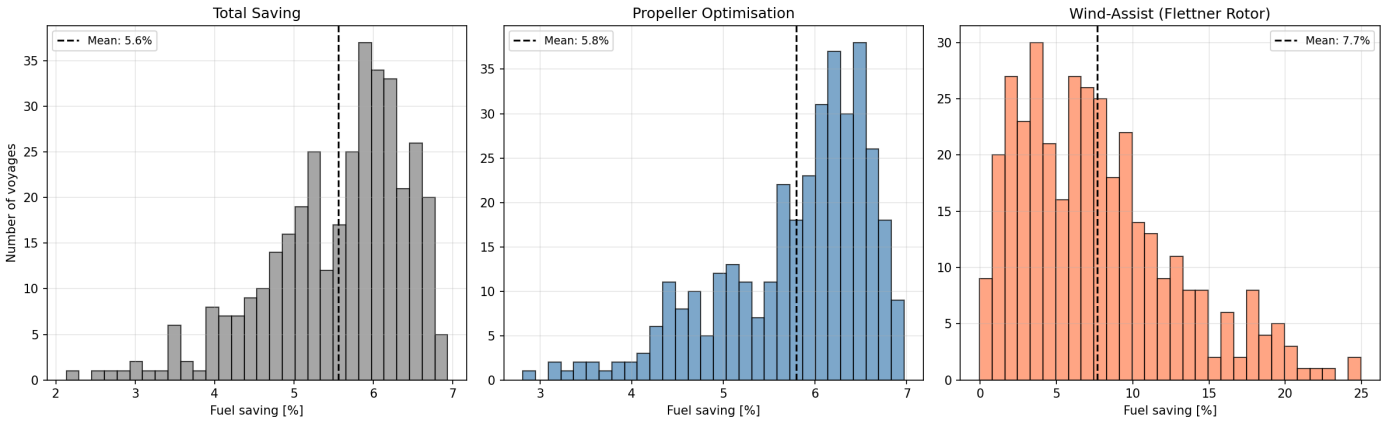

Figure 4-3: Distribution of per-voyage fuel savings. Left: total saving. Centre: propeller optimisation component. Right: wind-assist component.

4.4 Annual Cost Savings

At a fuel price of €650/tonne (MGO, typical for ECA zones):

Configuration	Annual fuel cost
Standard control, no rotor	€689 318
Standard control + Flettner rotor	€632 910
Optimiser, no rotor	€649 658
Optimiser + Flettner rotor	€597 794

Saving component	€/year	Percentage
Propeller optimisation	€39 660	5.8%
Wind-assist (Flettner)	€51 864	7.5%
Combined	**€91 524**	**13.3%**

4.5 Seasonal Variation

Season	Voyages	Std [kg/voy]	Opt+Fl [kg/voy]	Prop. opt.	Wind-assist
Q1 (Jan–Mar)	91	14 483	12 341	5.5%	9.5%
Q2 (Apr–Jun)	91	14 627	12 923	6.0%	5.8%
Q3 (Jul–Sep)	92	14 472	12 754	6.0%	5.9%
Q4 (Oct–Dec)	88	14 425	12 277	5.6%	9.6%

The Flettner rotor benefit roughly doubles in winter (Q1, Q4) compared to summer, reflecting the stronger and more consistent North Sea winds. The propeller optimisation saving remains stable at 5.5–6.0% year-round, as it depends on engine efficiency rather than weather.

4.6 Standard Control Limitations

The standard propeller control schedule was not designed to handle all possible sea conditions. In 7% of simulated voyages, the standard schedule drives the engine beyond its power limits in at least one hour – a situation where the operator would need to manually reduce speed.

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The Optimiser avoids this entirely by choosing pitch/RPM combinations that respect the engine's full operating envelope, maintaining the desired speed even in heavy weather.

5 Effect of Shaft Generator Operation

5.1 Why It Matters

The results above assume the vessel operates without a shaft generator (SG). Many vessels use an SG to supply electrical power from the main engine, avoiding the need for a separate diesel generator during transit. However, the SG introduces two constraints that affect propulsion efficiency:

1. **Additional engine load:** The SG extracts electrical power (assumed $150\,\mathrm{kW}$ average) from the main engine shaft, increasing fuel consumption.
2. **Minimum RPM constraint:** The SG must maintain its output frequency within $48\,\mathrm{Hz}$ to $60\,\mathrm{Hz}$, which sets a minimum engine RPM of approximately $640\,\mathrm{RPM}$ (shaft RPM $\geq 94.1\,\mathrm{RPM}$). Neither the standard schedule nor the Optimiser can operate below this RPM while the SG is online.

5.2 Results with Shaft Generator Active

Configuration	Fuel [tonnes/year]
Standard control, no rotor	1347.7
Standard control + Flettner rotor	1286.2
Optimiser, no rotor	1351.5
Optimiser + Flettner rotor	1290.6

Saving component	Tonnes/year	Percentage
Propeller optimisation	-3.9	-0.3%
Wind-assist (Flettner)	60.9	4.5%
Combined	**57.0**	**4.2%**

At MGO €650/tonne, the annual saving with SG active is €37 000/year (4.2%).

5.3 Why the Propeller Optimisation Saving Disappears

At $10\,\mathrm{kn}$ transit speed, the calm-water thrust demand requires only modest engine power. Without the SG, the Optimiser exploits this by running at low RPM and high pitch – well below the standard schedule's RPM – to reach the propeller's peak efficiency region. This accounts for the 5.8% saving.

With the SG online, the RPM floor forces both the standard schedule and the Optimiser to operate at near-identical shaft RPM ($\approx 94\,\mathrm{RPM}$). There is almost no room for the Optimiser to find a better operating point, and the pitch/RPM saving effectively vanishes.

5.4 Flettner Rotor Remains Effective

The Flettner rotor delivers the same absolute thrust reduction regardless of SG status – wind physics are unchanged. The relative saving appears smaller (4.5% vs 7.5%) only because the SG-loaded baseline consumption is higher ($1348\,\mathrm{t/yr}$ vs $1061\,\mathrm{t/yr}$).

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Season	Voyages	Std [kg/voy]	Opt+Fl [kg/voy]	Prop. opt.	Wind-assist
Q1 (Jan–Mar)	91	18 287	17 309	-0.2%	5.7%
Q2 (Apr–Jun)	91	18 616	18 022	-0.3%	3.5%
Q3 (Jul–Sep)	92	18 444	17 854	-0.3%	3.6%
Q4 (Oct–Dec)	88	18 368	17 402	-0.3%	5.6%

5.5 Side-by-Side Summary

Metric	SG offline	SG online (150 kW)
Baseline fuel [t/yr]	1060.5	1347.7
Propeller optimisation saving	5.8% (61 t/yr)	-0.3% (-4 t/yr)
Wind-assist saving	7.5% (80 t/yr)	4.5% (61 t/yr)
Combined saving	13.3% (141 t/yr)	4.2% (57 t/yr)
Annual cost saving	€92k	€37k

The SG operating mode has a substantial impact on the achievable savings. When the SG is offline, the Optimiser delivers its full 5.8% benefit. When the SG is online, the RPM constraint removes this degree of freedom, and only the Flettner rotor's 4.5% wind-assist benefit remains.

At higher transit speeds, the propeller optimisation saving recovers under SG constraints because the thrust demand pushes the engine above the SG RPM floor:

Transit speed	Propeller opt.	Wind-assist	Combined
$10\,\mathrm{kn}$	-0.3%	4.5%	4.2%
$12\,\mathrm{kn}$	3.3%	4.4%	7.7%
$14\,\mathrm{kn}$	4.9%	4.2%	9.1%

5.6 Shaft Generator vs Auxiliary Genset

An important question is whether the shaft generator is the most fuel-efficient way to supply $150\,\mathrm{kW}$ of electrical power at $10\,\mathrm{kn}$.

With the SG, the baseline fuel rises from $1061\,\mathrm{t/yr}$ to $1348\,\mathrm{t/yr}$ — an increase of $287\,\mathrm{t/yr}$ for $150\,\mathrm{kW}$ over $7439\,\mathrm{h/yr}$. This corresponds to an *effective* specific fuel consumption of $257\,\mathrm{g/kWh}$, well above the 210–$230\,\mathrm{g/kWh}$ of a typical auxiliary diesel genset.

The reason is the SG RPM floor: at $10\,\mathrm{kn}$ the engine would naturally operate at lower RPM with better fuel efficiency, but the SG forces it to $640\,\mathrm{RPM}$ minimum, shifting operation to a less efficient part of the fuel map.

At a typical genset SFOC of $215\,\mathrm{g/kWh}$, a separate auxiliary genset would save approximately $47\,\mathrm{t/yr}$ (€31 000/year) in fuel compared to the shaft generator. This saving is in addition to the Optimiser and Flettner benefits.

Figure 5-1 illustrates this contrast across the full year.

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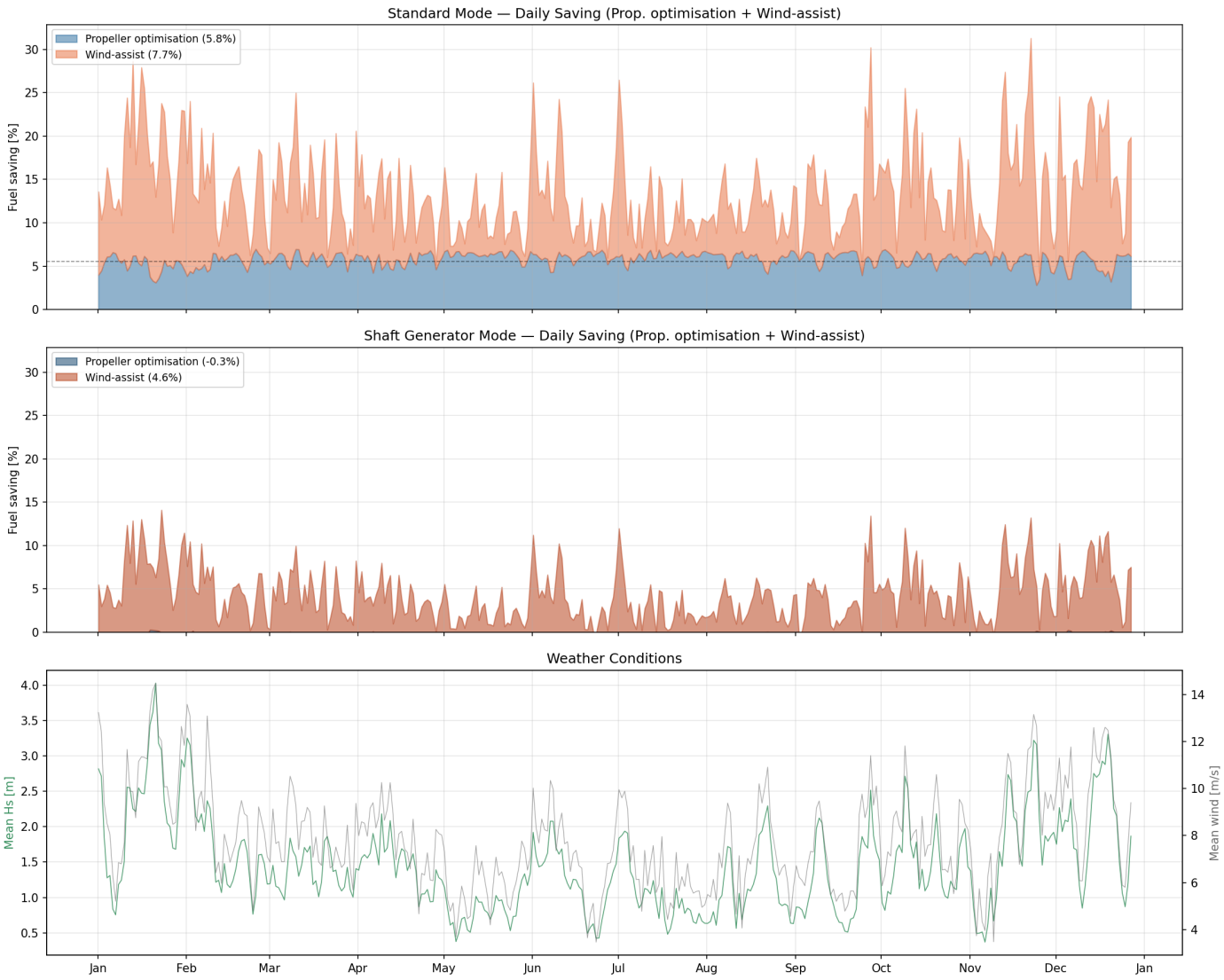

Figure 5-1: Daily fuel savings comparison: standard mode (top panel, SG offline) vs shaft generator mode (bottom panel, SG online at 150 kW). The propeller optimisation saving (blue) collapses in SG mode, while the wind-assist saving (orange) persists.

6 Conclusions

The simulation of 362 round-trip voyages on the Rotterdam–Gothenburg route using actual 2024 weather data demonstrates fuel saving potential from two complementary technologies. The achievable savings depend on whether the shaft generator is in use:

Without shaft generator (SG offline):

1. **The Optimiser** saves **5.8%** of fuel (61 tonnes/year, €40 000/year). This is a consistent, weather-independent benefit delivered on every voyage by running the engine at a more efficient operating point.
2. **Flettner rotor wind-assist** saves an additional **7.5%** (80 tonnes/year, €52 000/year), varying with wind conditions – roughly 6% in summer and 10% in winter.
3. **Combined saving: 13.3%** (141 tonnes/year, €92 000/year).

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With shaft generator (SG online, 150 kW, 48–60 Hz):

4. The SG RPM floor removes the Optimiser's freedom to select lower RPM, reducing its pitch/RPM saving to effectively zero (-0.3%). At higher transit speeds ($12-$-$14\,\mathrm{kn}$), the saving recovers to 3.3–4.9% as thrust demand pushes the engine above the SG RPM floor.
5. The **Flettner rotor remains effective at 4.5%** (61 tonnes/year). The lower percentage reflects the higher baseline consumption from the SG parasitic load, not a reduction in actual wind-assist performance.
6. **Combined saving with SG: 4.2%** (57 tonnes/year, €37 000/year).
7. **Shaft generator vs auxiliary genset:** At $10\,\mathrm{kn}$, the effective fuel cost of generating $150\,\mathrm{kW}$ via the shaft generator is $257\,\mathrm{g/kWh}$ — significantly higher than a typical auxiliary genset at $215\,\mathrm{g/kWh}$. A separate genset would save approximately €31 000/year.

In both modes:

7. The Optimiser eliminates engine overload situations that occur with the standard schedule in heavy weather, improving operational reliability.

These results assume clean hull and propeller surfaces. Fouling (biological growth on the hull and propeller) increases resistance over time, which would increase the absolute fuel savings from both technologies while also increasing total consumption. The simulation tool supports fouling scenarios and can provide estimates for any assumed maintenance interval.

7 Appendix: Methodology Notes

The physical models underlying the simulation are based on established naval architecture methods:

- **Hull resistance:** Interpolated from vessel-specific service prediction data (model tests).
- **Wave added resistance:** Strip-theory hydrodynamic transfer functions (PdStrip) integrated over JONSWAP wave spectra.
- **Wind resistance:** Blendermann (1994) method for general cargo vessels.
- **Flettner rotor:** Lift/drag coefficients from Bordogna et al. (2019) with aspect-ratio correction; rotor drive power from Theodorsen & Regier (1944).
- **Fouling:** Hull roughness model of Townsin & Dey (2003); propeller blade roughness from Mosaad (1986).
- **Weather data:** NORA3 hindcast archive (MET Norway), $3\,\mathrm{km}$ resolution, hourly time steps.
- **Propulsion model:** C-series propeller model with full pitch/RPM optimisation against the MAN L27/38 engine envelope.

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